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# ***Alarm Surveillance***

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## ***Chapter 1***

This chapter is designed to provide the student with the skills needed to operate the first line of support in the NMS

### **OBJECTIVES:**

Upon completion of this chapter the student will be able to:

- Detail the architecture of the Fault Manager and describe its main functionality
- Detail Alarm Handling
- Associating/Creating Trouble Tickets
- Understand Alarm Filtering
- Detail the main functionality of the Iconic Map
- Using on-line documentation

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# 1 Alarm Surveillance

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## FAULT MANAGEMENT – NS-T

### OVERVIEW OF SYSTEM ARCHITECTURE

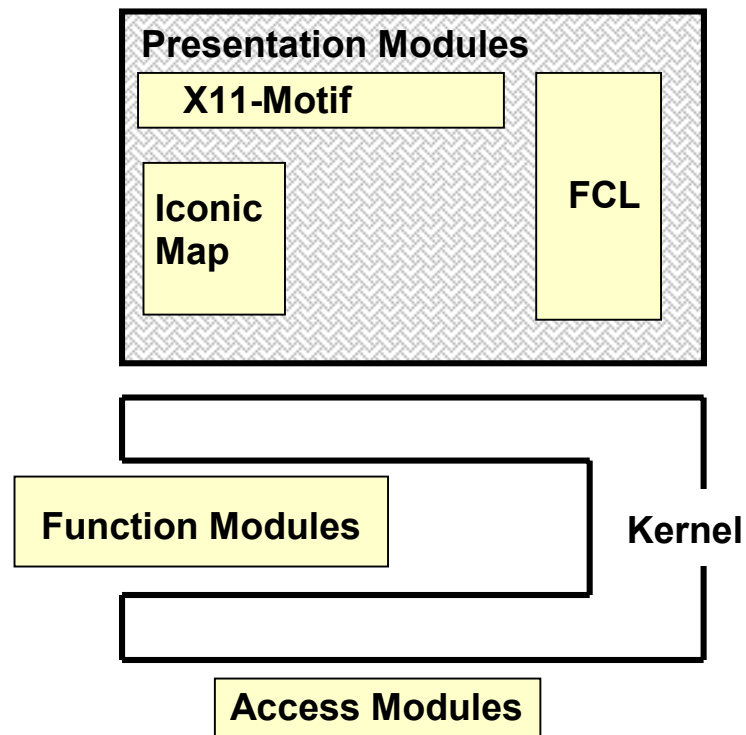


Figure 1.1 NS-T Components

## OVERVIEW OF SYSTEM ARCHITECTURE

### NS-T COMPONENTS

NS-T is an integrated software package made up of several components. These components provide a solid, modular, distributed architecture and an open development environment that complies with OSI (Open Systems Interconnection) Standards.

These components are:

- TeMIP Framework
- TeMIP Fault Management
- Development Toolkits:
  - Framework Developer's Toolkit
  - Visual TeMIP Toolkit
  - SNMP Toolkit
  - OSI AM Toolkit
  - ASCII AM Toolkit
  - Corba Toolkit

The distributed architecture includes presentation, function and access modules and the capability to integrate existing applications into an existing Network Management System. In addition, NS-T allows the user to connect to networks running under different protocols and to monitor and control the network elements.

## OBJECT ORIENTED SYSTEM

NS-T is an Object Oriented System. As the name *object-oriented* implies, *objects* are key to understanding object-oriented technology. Examples of real-world objects: your desk, a dog and your bicycle.

These real-world objects share two characteristics: they all have *state* and they all have *behaviour*. For example, dogs have state (colour, breed, hungry) and dogs have behaviour (barking and fetching). Bicycles have state (current gear, two wheels, number of gears) and behaviour (braking, accelerating, changing gears).

Software objects are modelled after real world objects in that they, too, have state and behaviour. A software object maintains its state in *variables* and implements its behaviour with *methods*.

You can represent real-world objects using software objects. You might want to represent real-world dogs as software objects in an animation program or a real-world bicycle as a software object within an electronic exercise bike. However, you can also use software objects to model abstract concepts. For example, an event is a common object used in GUI window systems to represent the action of a user pressing a mouse button or a key on the keyboard.

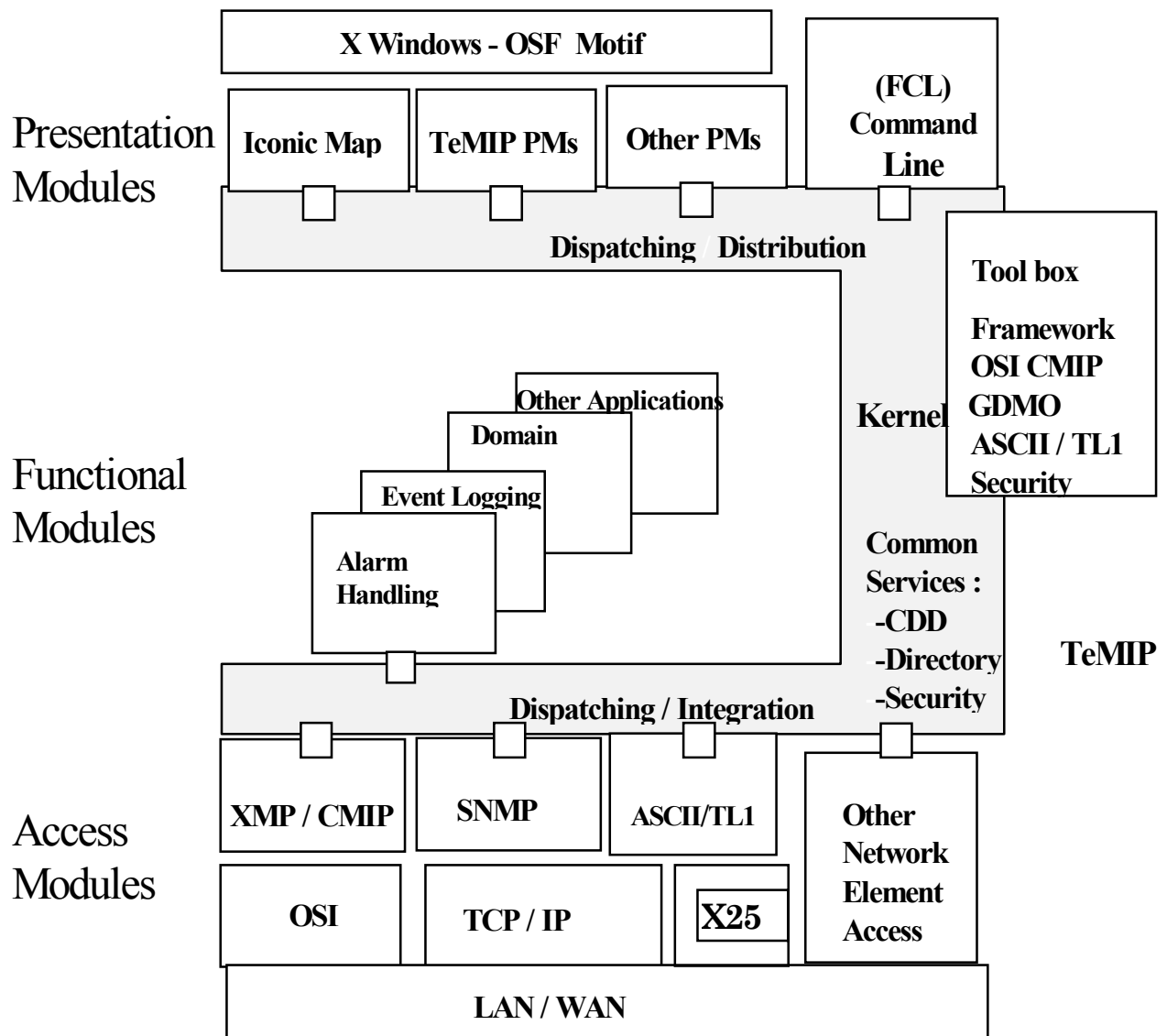


Figure 1.2 NS-T Framework



## NS-T Framework

NS-T Framework provides a base management system, which consists of a set of three interactive management modules that provide their own individual functionality.

These modules are as follows:

### Access Modules

Access Modules provide access to and information about different classes of managed objects (Network Elements).

### Function Modules

Function Modules provide value-added management functions for the managed objects.

### Presentation Modules

Presentation Modules provide a consistent user interface for management operations and user access to managed objects.

The NS-T Kernel provides the system services that enable the three types of management modules to be enrolled and communicate with each other. It defines the Applications Programming Interface (API) that allows modules to invoke each other's services. It provides access to the Management Information Repository (MIR), which is an object-oriented database for the storage of management information.

## Access Modules

The NS-T Access Modules (AMs) communicate with the components it manages using a data driven object model that allows new attributes and devices to be easily added without programming. Four Access Modules talk to the devices they represent using native protocols, including ASCII, SNMP, CORBA and CMIP - as do most network management solutions. The NS-T Access Modules, however, go a step further. Besides filtering and prioritizing events, the Access Modules also convert the device specific messages into a standard format. By having all the messages in the framework based on a single format (ISO-based message sets), the object-oriented system only needs the messages to be defined once to allow all the modules to communicate effectively. The Framework is able to support thousands of events per second while providing all the services because the Access Modules are running in a threaded environment.

Ericsson has created many customized access modules for NMS.

## Function Modules

NS-T Fault Management provides additional management functions, which have been tailored specifically for the telecommunications environment and OSI.

Function Modules (FMs), which are nothing more than applications, take advantage of the powerful object-oriented infrastructure to provide unique capabilities for the management areas of event logging, trouble ticketing, domain management, configuration and accounting. Examples of Function modules include:

- Alarms FM
- Notification FM
- Trouble Ticketing (via Resolution)
- TeMIP Expert (rules engine)
- Domain FM
- Registration FM
- Event Filtering and Correlation FM

Because the NS-T Framework is object oriented, Presentation Modules can directly access the Function Modules to display user appropriate information. Function Modules can also interact with each other, allowing them to be used as building blocks to easily construct and manage complex environments.

## Presentation Modules

Presentation Modules (PMs) support a director's user interface requirements and provide clear and consistent directives that are performed against a diverse population of managed entities. NS-T security, audit trail logging, and distribution control the viewpoint provided by the PM. Regardless of the mechanism utilized, Motif, Windows NT client, Java, CORBA, or OSI Q3, the Presentation Module must make use of the same consistent API interface.

## WHAT DOES NS-T LOOK LIKE?

### Windows and Views

End user access to the NS-T functions includes a graphical interface based on DIGITAL UNIX/Motif[R] and the X-Windows system. This provides a high-quality display on which the NS-T management operations can be performed quickly and easily through multiple windows, menu-driven selections, and a comprehensive set of user commands.

**The Iconic Map window** shows the selected part of the management domain (see the later section).

**The Alarm Handling Real-Time window** shows the alarm summary and list of Not-Closed alarms (see the later section).

**The Console window** displays information messages. In certain cases, error messages are also displayed in this window.

**The Desktop window** has icons that represent the windows displayed. The Desktop window allows the user to dismiss windows and remove them from the display without closing the application.

### Consistent Management View

The NS-T management interface provides a consistent view of the management environment, and a consistent means of interacting with that environment, regardless of the different types of network element or protocols that may be involved.

### Multiple Screen Support

If there is more than one screen available, NS-T displays the Iconic Map and the NS-T Fault Management windows on different screens.

## NS-T User Interfaces

NS-T provides two main user interfaces that allow the user to use multiple presentation devices and styles.

Each user interface is implemented as a Presentation Module (PM), and provides a consistent way of performing functions and gaining access to entities.

The two main user interfaces are:

The **Iconic Map PM**, which provides a X/Motif environment.

The **Framework Command Line PM**, which provides a character-based command line environment.

## Starting User Interfaces

**Iconic Map** - To start the Iconic Map, enter the following command:

**%mcc\_iconic\_map**

**Framework Command Line (FCL)** - To start FCL, enter one of the following commands:

**%manage** or **%mcc\_fcl\_pm**

The user can have the Iconic Map and FCL active at the same time.

When the user starts the Iconic Map, they will also see the Console and Desktop Manager windows.

- The **Console Window** displays information messages, including error messages.

Messages displayed in the *Console* window consist of a message number, day of the week, month, date and time, followed by the name of the window that generated the message and the message text. The format is as follows:

*Message number (001-999) - Day of Week - Month - Date of month - Time in hh:mm:ss - Window Name- Message text*

Use this window to diagnose problems with the operations the user performs. Messages in different categories are displayed in different colors. The default display is black for information, green for warning, and red for error. The user can customize the colors by changing the `mcc_resource.dat` file.

- The **Desktop Manager** contains icons that represent the windows that are displayed on the screen. It allows the user to dismiss windows and remove them from the display without closing the application.

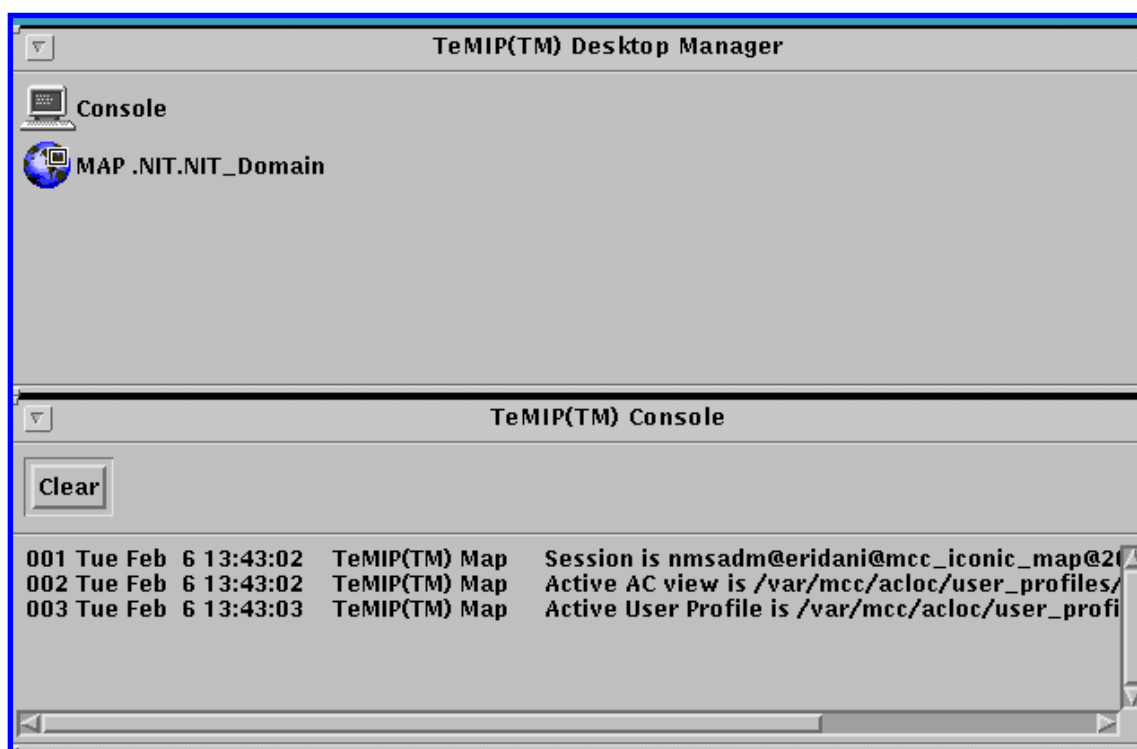


Figure 1.3 Desktop Manager and Console Windows

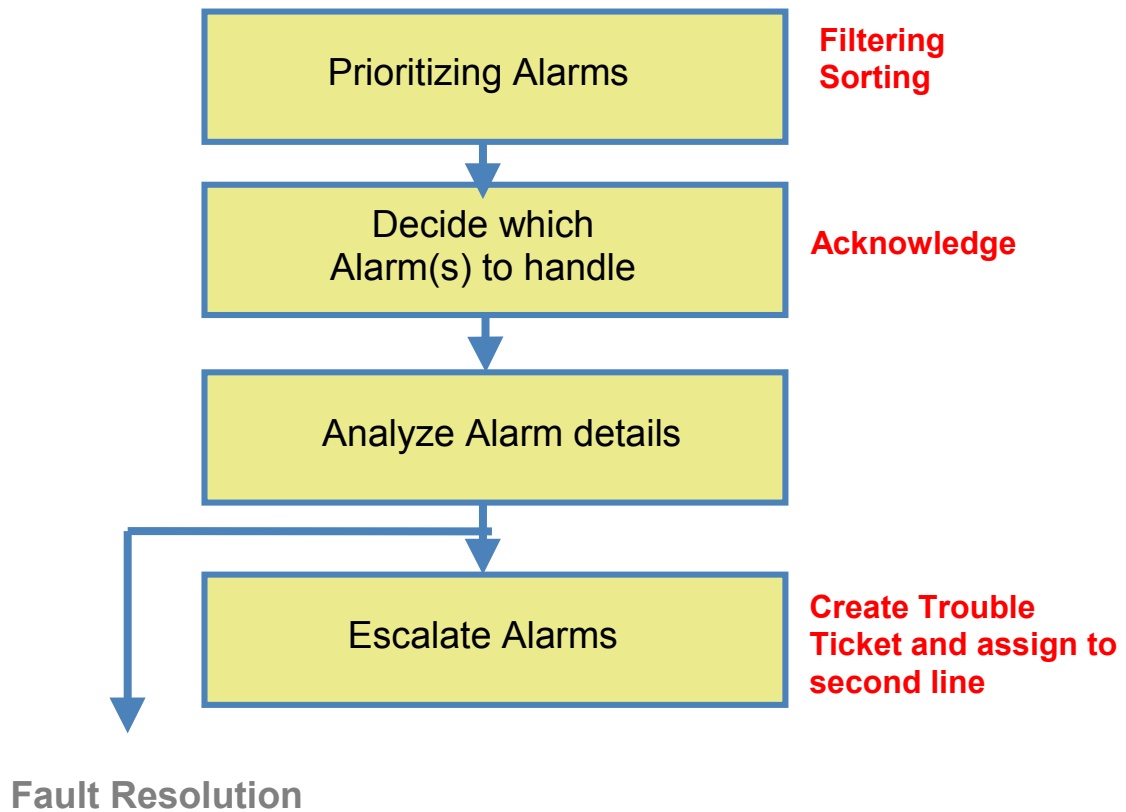


Figure 1.4 The Alarm Surveillance Operator Processes

## OVERVIEW OF THE ALARM HANDLING WINDOW

### ***BASIC CONCEPTS AND FUNCTIONALITY***

The handling of alarms in Ns-T is a process that involves cooperation between the following FMs:

- Alarm Handling FM
- Alarms FM
- Notification FM

The NS-T Alarm Handling FM provides handling of OSI-formatted alarm information generated by network elements. In effect, the NS-T Alarm Handling FM specifically follows the ISO standards for network problem management systems. By following these standards, NS-T provides the essential basis for the development of problem management systems in an OSI environment.

The Alarms FM provides a basic alarm generation mechanism based on alarm rule expressions. When alarm rule conditions are met, an alarm is triggered and the user is notified. The Alarms FM can also execute a script to perform any function the user defines.

The collection and presentation of alarm information is handled in real-time, enabling Operators to react quickly to problems in the network, identify the source of the malfunction as quickly as possible, and if necessary, implement maintenance activity.

The Notification FM provides the alarm monitoring, collection and notification capabilities, which are used by the Alarm Handling FM to present the alarm information to the user interface as an OSI-compliant alarm report. The Notification FM provides services for reporting alarms and events received through the Access Modules or Alarms FM to other management modules, including the NS-T Alarm Handling and Event Logging FMs. When an alarm condition is met, the Notification FM consults the Domain FM to check that the entity generating the alarm is in fact a member of the domain in question and then the user is notified by an alarm on the screen.

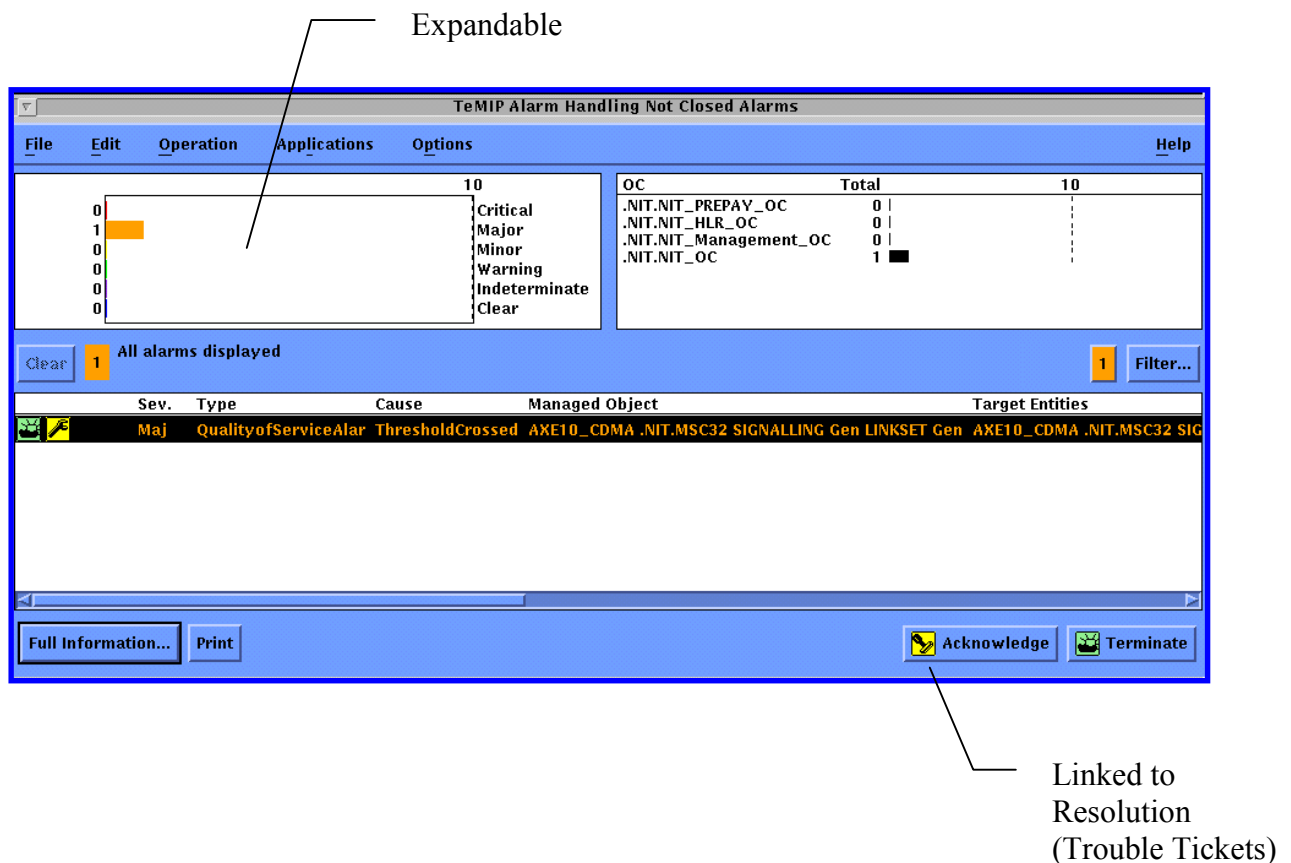


Figure 1.5 – Alarm Handling Window

## FEATURES OF THE ALARM HANDLING FM

NS-T Alarm Handling FM allow users and system administrators to:

- Create different views of the management domain for different aspects of problem management
- Filter alarm data according to an OSI conforming discriminator, to monitor specific events
- Schedule the alarm handling function to be automatically activated at specific intervals, for specified periods
- Convert filtered alarm data into OSI-formatted alarm objects, to allow handling and maintenance of the data
- Store the alarm objects in a repository



- Notify in real-time when an alarm object has been created or its state changes
- Manage the alarm objects and the alarm handling environment
- Escalate the severity level for alarms that are not acknowledged within a specified time
- Create and assign Trouble Ticketing Reports (TTRs)
- Associate specific alarms with TTRs
- Search for and retrieve alarm data for viewing or printing
- Archive alarm data in an external database for off-line processing
- Access, monitor and manage alarm information stored in a remote alarm repository (available in a distributed systems)

## OPERATION CONTEXTS

NS-T Alarm Handling is based on **Operation Contexts (OCs)** and **Alarm Objects (AOs)**. Operation contexts control the way that alarm collection is carried out; alarm objects enable the user to handle and manipulate alarm information through the NS-T management interface.

In order to properly understand how to use NS-T Alarm Handling, it is essential to fully understand the concept of the operation context.

In any problem management scheme, the network alarm information can be used to achieve different management objectives. Different types of alarm data are required for different reasons, each representing one aspect of network activity of interest to a particular user.

In NS-T, the **operation context** is used to define a particular view of alarm activity in a management domain (or domain tree). By assigning appropriate values to the available arguments when an operation context is created, the user defines how the alarm handling feature is to behave to fulfill some aspect of problem management.

The operation context represents the **scope** of user interest in network alarm activity. The operation context is an extremely flexible mechanism, since the user can define the scope of interest to be any alarm received for the associated domain, right down to a very specific type of alarm, of a particular severity,

probable cause, for a particular managed object and so on. The scope could be based on geography, equipment type, or any other combination of available arguments and attributes.

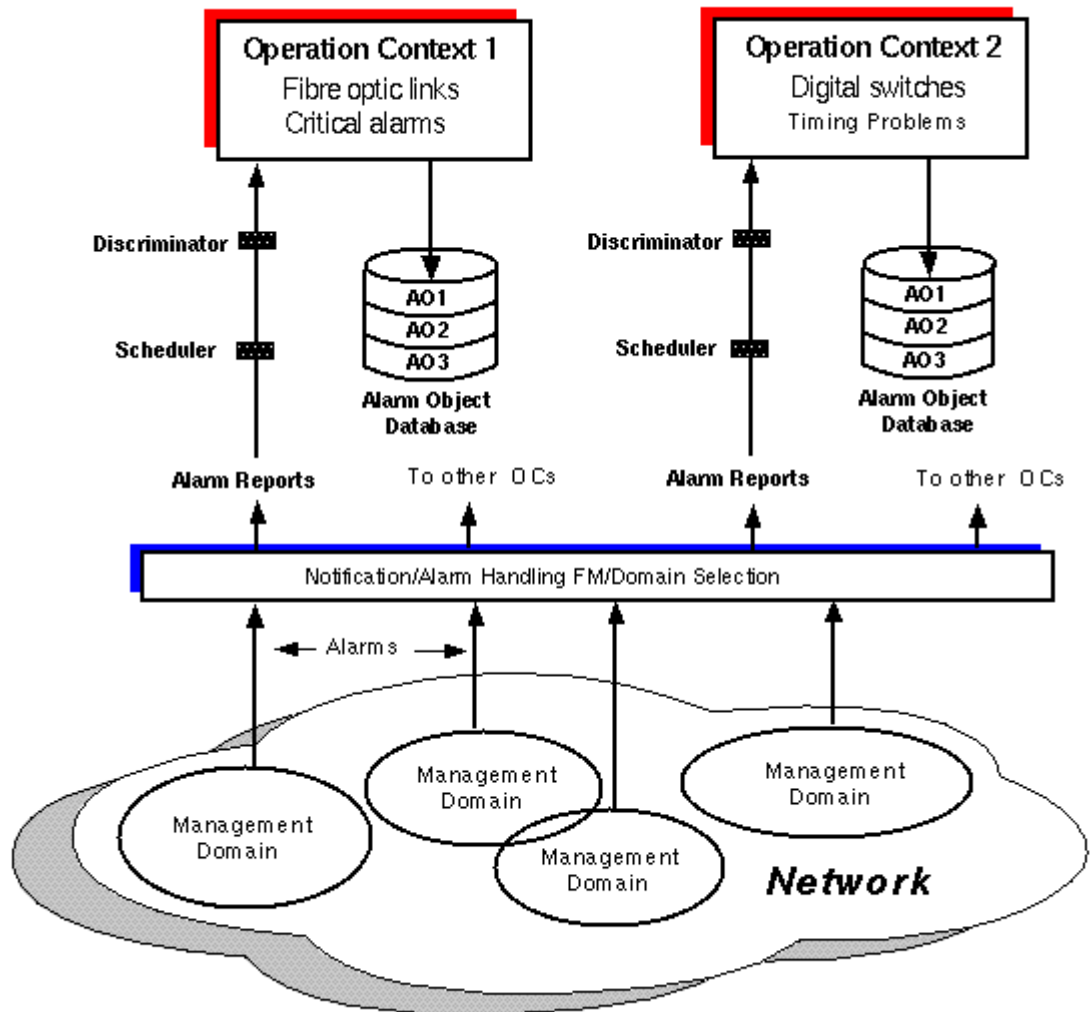


Figure 1.6 – Operation Context

Figure 1.6 illustrates the principle, where alarm information from the same domain is handled differently by the Alarm Handling FM for two different operation contexts. The operation contexts carry the alarm databases.

For example, two of the operation contexts could be defined as follows:

**Operation Context 1:**

Collect only alarms of severity level *critical* from all optical fiber communication links in a certain geographic area.  
Scheduling not required.

**Operation Context 2:**

Collect timing problem alarms from the digital switch in a specified main exchange. Scheduling set to activate alarm handling from midnight to 8.00am each day.

## USER TASKS

NS-T will carry out its alarm collection function according to how it is configured for the user's particular fault management requirements. NS-T users should know how to:

- View Alarms
- Check an alarm's "Full Information"
- Acknowledge a new alarm
- Terminate an alarm
- Clear a filter
- Create a Trouble Ticket Report (TTR)
- Associate an alarm with an existing TTR
- Create High Level Filters

Users should also be aware that alarms can be cleared by the System Administrator.

Along with the Iconic Map, the alarm-handling window is the most important view of the NMS. It is used to monitor alarms from the network and take the necessary action. The configuration should be set to show alarms from the Operation Contexts in the network that are set to monitor.

Alarm collection is a background process and does not rely on any NS-T user being logged on to the system. By enabling the Alarm Handling function, the user enables the display of alarms.

## View Alarms

There are two ways the Alarm Handling Window can be opened from the Iconic Map. Either Click on the Icon named NS-T Alarm Handling, or use the menu:

### /application/TeMIP Alarm Handling

The major part of the Alarm Handling window contains a list of alarms from the Operation Contexts that the user has set to monitor and that have passed through the filtering mechanisms of the Operation Context.

Above the alarm list, to the left, is the total number of alarms of each severity.

Above the alarm list, to the right, is the number of alarms in each Operation Context.

From the Alarm Handling window, the user manages all alarms. As an object, the alarm can be managed and acted upon in various ways by the user until the network event that generated the alarm has been investigated and resolved. At this point the alarm object becomes an item of historical data.

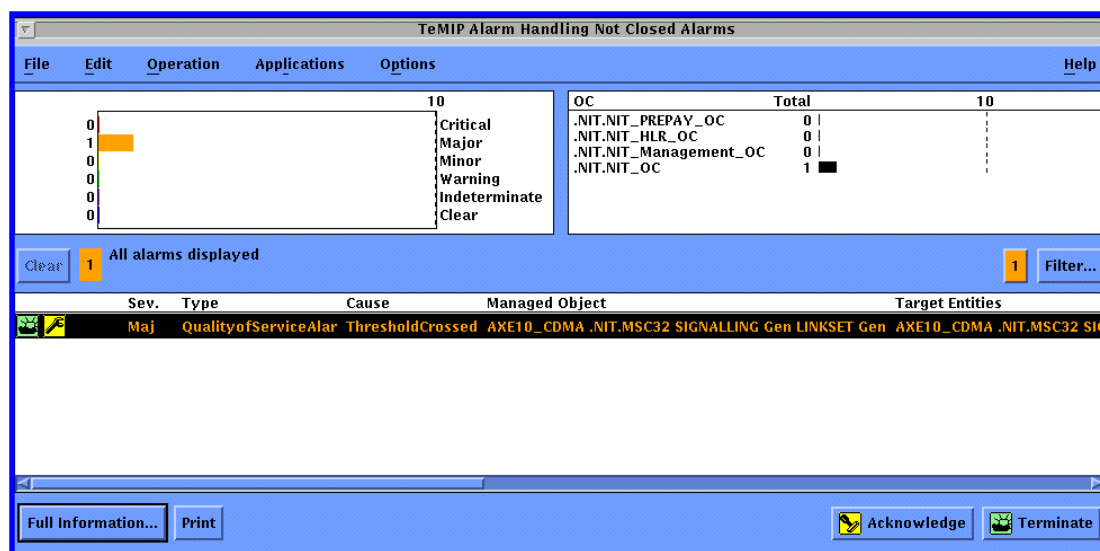


Figure 1.7 – Alarm handling window with a selected alarm

| Alarm Information : OPERATION_CONTEXT.NIT.NIT_OC alarm_object 105679                 |                                                                      |
|--------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Attribute Name                                                                       | Attribute Value                                                      |
| Characteristics                                                                      | AXE10_CDMA [REDACTED]_NS::NIT.MSC32 SIGNALLING Gen LINKSET Gen       |
| Managed Object                                                                       | AXE10_CDMA [REDACTED]_NS::NIT.MSC32 SIGNALLING Gen LINKSET Gen       |
| Target Entities                                                                      | QualityofServiceAlarm                                                |
| Alarm Type                                                                           | ThresholdCrossed                                                     |
| Event Time                                                                           | Thu Jan 11 12:55:30                                                  |
| Probable Cause                                                                       | CONGESTION IN ALARM SYSTEM                                           |
| Specific Problems                                                                    | 41                                                                   |
| Notification Identifier                                                              | Domain [REDACTED]_NS::NIT.NIT_Domain                                 |
| Domain                                                                               | Major                                                                |
| Perceived Severity                                                                   | *** ALARM 041 A2/APT ***MT52 3000*CV100***U 010111 1255              |
| Additional Text                                                                      | CCITT7 LINK SET SUPERVISION<br>LS SPID NACTSL<br>2-457 STG 0<br>END" |
| Original Severity                                                                    | Major                                                                |
| Original Event Time                                                                  | Thu Jan 11 12:55:30                                                  |
| Status                                                                               | Terminated                                                           |
| State                                                                                | Handled                                                              |
| Problem Status                                                                       | False                                                                |
| Clearance Report Flag                                                                | False                                                                |
| Escalated Alarm                                                                      | "temip"                                                              |
| Acknowledgement User Identifier                                                      | MMIS [REDACTED]_NS::mmis1 TT "0000000003"                            |
| Handled By                                                                           | "temip"                                                              |
| Handled User Identifier                                                              | "root"                                                               |
| Termination User Identifier                                                          | Mon Jan 29 10:48:31                                                  |
| Acknowledgement Time Stamp                                                           | Mon Jan 29 10:48:31                                                  |
| Handle Time Stamp                                                                    | Tue Feb 6 12:39:50                                                   |
| Termination Time Stamp                                                               | Thu Jan 11 14:25:02                                                  |
| Creation Timestamp                                                                   |                                                                      |
| Counters                                                                             |                                                                      |
| Problem Occurrences                                                                  | 1                                                                    |
| Critical Problem Occurrences                                                         | 0                                                                    |
| Major Problem Occurrences                                                            | 1                                                                    |
| Minor Problem Occurrences                                                            | 0                                                                    |
| Warning Problem Occurrences                                                          | 0                                                                    |
| Indeterminate Problem Occurrences                                                    | 0                                                                    |
| Clear Problem Occurrences                                                            | 0                                                                    |
| SA Total                                                                             | 0                                                                    |
| <div> <div>Print</div> <div>Previous</div> <div>Next</div> <div>Dismiss</div> </div> |                                                                      |

Figure 1.8 – Full information on a selected alarm

### Check an Alarm's Full Information

An alarm contains approximately 30-40 different attributes. Not all of them can be seen directly in the alarm list. To see them all, double click on the actual alarm or click on the **Full Information** button at the bottom of the Alarm Handling window.

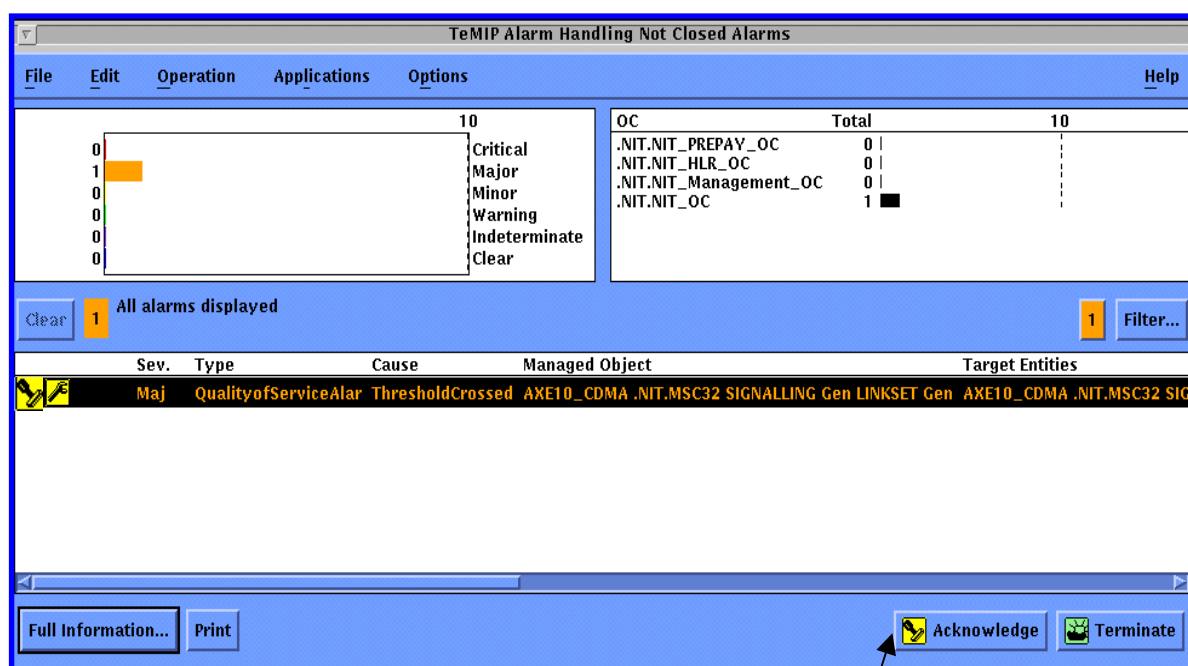
If the user clicks on an alarm, it appears in reverse video. This means that it can have different colors due to the settings of alarm colors.

## Acknowledge a new alarm

When a new alarm appears in the alarm list in the Alarm Handling window, it does so at the top or at the bottom of the list, due to configuration. A new alarm has to be acknowledged by a user.

To acknowledge one or multiple alarms, choose the alarm(s) from the list, and click on **acknowledge** to the right at the lower part of the window.

When an alarm is acknowledged a stamp symbol appears in the alarm list. This symbol replaces the light bulb symbol that represents an unacknowledged alarm. The user's user name is also added to the Full Information window.



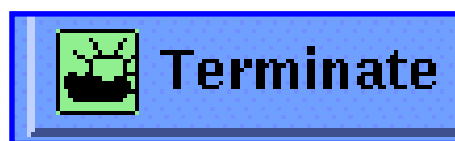
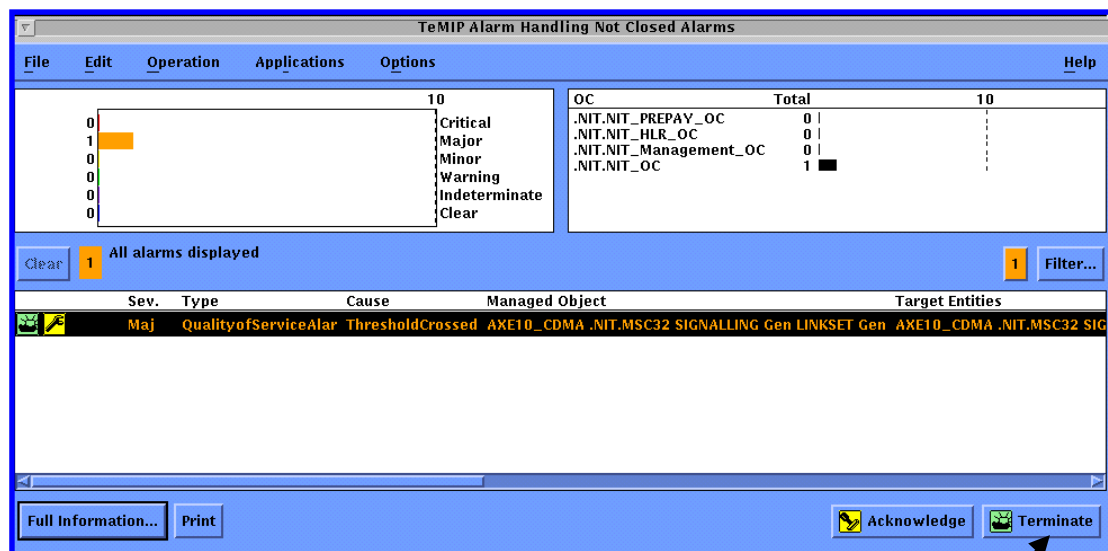
Selecting an alarm and clicking on the Acknowledge button will place a stamp next to the alarm.

Figure 1.9 Acknowledging an alarm

## Terminate an Alarm

When an alarm has been handled according to the policies set by the System Administrator, it has to be removed from the list. This is done by using the terminate function. Simply select the alarm the user wish to terminate and press the terminate button at the lower right corner of the window.

Note that terminating an alarm only removes it from the alarm list. The alarm is still available in the alarm database, until it is purged or archived according to the user's company's policies.



Selecting an alarm and clicking on the Terminate button will place the terminated symbol next to the alarm.

Figure 1.10 Terminating an alarm

## Sorting Alarms

Alarms in the Not-Closed Alarm Handling Window are presented in either ascending or descending order and according to the criteria selected. Click Mouse Button 1 on the Perceived Severity field in the header to change the direction of the arrow head. This changes the sorting from descending to ascending order (only for this field). You can change the sorting of any of the displayed fields in this way.

| AD Managed O... | A | A | AD Perceived Se... | AD Event Time | AD Probable Ca... |
|-----------------|---|---|--------------------|---------------|-------------------|
| OPERATIO...     |   |   | Critical           | 11/16/2000... | StorageCap...     |
| OSI_SYSTE...    |   |   | Major              | 10/11/2000... | CallEstablis...   |
| BSS_demo...     |   |   | Minor              | 09/26/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Minor              | 10/11/2000... | CallEstablis...   |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |
| OSI_SYSTE...    |   |   | Warning            | 11/15/2000... | BandwidthR...     |

Filtered Alarms (Total): 206      Filtered Alarms (New): 206

Figure 1.11 Alarm sorting: Perceived severity descending



## Create a Trouble Ticketing Report

It is possible to launch an empty TT schema from the Iconic Map **or** from the Alarm Handling application.

It is launched from the Iconic Map by selecting **Applications/Resolution Trouble Ticketing** from the menu bar or by a single click on the Resolution Trouble Ticketing icon in the icon bar.

Selecting Applications/Resolution Trouble Ticketing from the menu bar launches Trouble Ticketing.

All Trouble Ticket management starts with a new network alarm that is displayed in the Alarm List.

If a similar alarm has been handled before, the alarm should be associated to an existing TT by using the TT-Associate application. If not, a new TT should be created by using the TT-Create application.

When an alarm has been handled, i.e. has a connected TTR, the TTR can easily be modified by launching the Resolution TT schema via the TT-Modify application.

To create a Trouble Ticket do the following:

1. Open the NS-T Alarm Handling (Alarm List) from the Iconic Map either by selecting Applications/NS-T Alarm Handling from the menu bar or by clicking on the NS-T Alarm Handling icon in the icon bar.
2. Select one or several not handled alarm(s) from the Alarm List (The TT-Create application cannot be opened if no alarm is selected)

**Note: It is recommended to not select more than 50 alarms in Alarm List.**

3. Launch the TT-Create application from either Operation/Trouble Ticket/Create in the menu bar, as in the Figure 2. below, or from the popup menu in the Alarm List by selecting Trouble Ticket/Create.

One of the two available menus for launching TT applications from the Alarm Handling application. The other is activated by pressing the right mouse button in the Alarm List.

Starting the application, the user may experience a few seconds delay while the application is loading before the window is displayed.

4. Fill in the mandatory fields. This will be covered in more detail in the Trouble Management (Resolution) module.

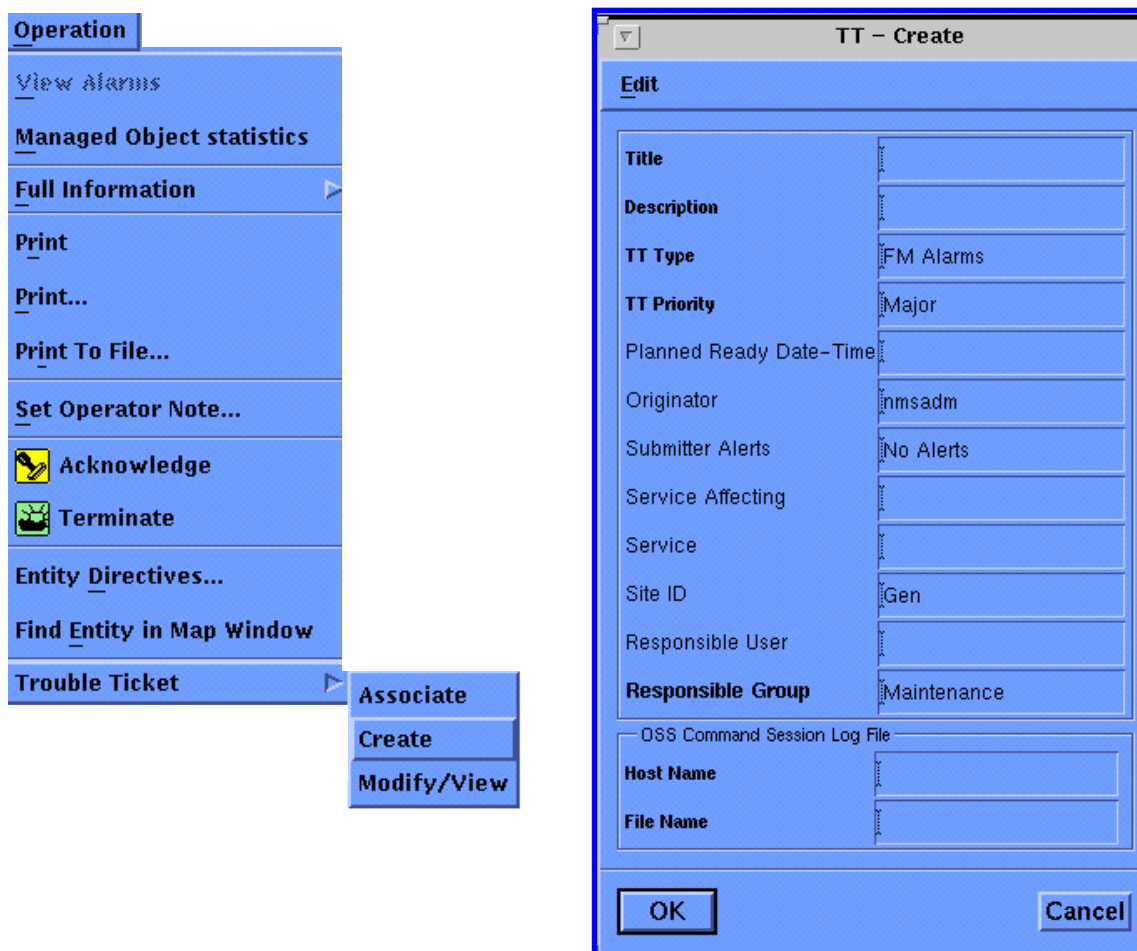


Figure 1.12 - Creating a Trouble Ticket Report

## Associate an Alarm with an Existing Trouble Ticket Report

The purpose with the associate feature is to give the user an opportunity to add more alarms to an existing TTR.

1. Open the NS-T Alarm Handling (Alarm List) from the Iconic Map either by selecting **Applications/NS-T Alarm Handling** from the menu bar or by clicking on the **NS-T Alarm Handling** icon in the icon bar.

2. Select one or several not handled alarm(s) from the Alarm List (The TT-Associate application cannot be opened if no alarm is selected)

**Note: It is not recommended to select more than 50 alarms in the Alarm List. If more than 50 alarms must be associated to a TT, one or more consecutive associations should be made.**

3. Launch the TT-Associate application from either **Operation/Trouble Ticket/Associate** in the menu bar, or from the popup menu in the Alarm List by selecting **Trouble Ticket/Associate**.

The TT Associate window opened when launching the associate application from the menu bar or from the popup menu in the Alarm List.

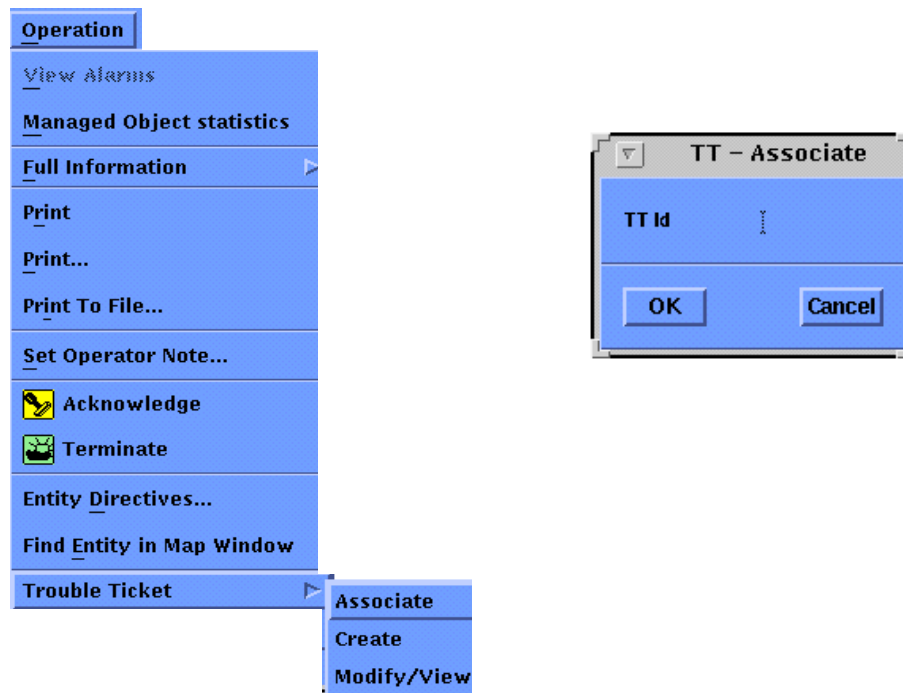
4. Fill up the TT Identifier field. No popup menu is connected to this field.

5. Submit the information by pressing the **OK** button and wait until the association has been performed, i.e. a message box is displayed as below. This Information window appears after a successful association.

6. Finish the associate session by pressing the OK button in the TT-Information window. If the association is successful, a number of actions are performed:

- The Trouble Ticket Id is written into the "Handled By" attribute of the selected alarm(s)
- The Problem Status icon in the Alarm List is changed from a bolt (Not Handled) to a wrench (Handled)
- If the alarm(s) have are not already acknowledged, the Alarm State icon in the Alarm List is changed from a bulb (Outstanding Alarm) to a stamp (Acknowledged)

NS-T has the ability to filter alarms so that duplicate alarms can be seen as one alarm.



*Figure 1.13 – Associating an alarm with an existing Trouble Ticketing Report*

If an error occurs while running the TT Associate application, an error message window will be displayed and no changes will be propagated to the selected alarm(s).

After a successful operation the popup might contain some additional information about the operation status. The extra information is separated from the success message with a dashed line.

In order to associate alarms to an already created trouble ticket:

- Select one or several alarm(s) from the Alarm List
- Launch the TT Associate application from either the menu bar or from the popup menu
- Fill up the TT Identifier field

- Submit the information by pressing the OK button
- Finish the session by pressing the OK button in the TTR Information window

## **TYPES OF FILTERING**

In NS-T, Event Filtering is carried out at two levels, a **low level** and a **high level**.

### **Low Level Filtering** (System Administrator task)

Low level filtering is the functionality that filters events at **source** using the discriminator construct. Low level filtering can be used to prevent events reaching the Operator that are generated as a result of another, perhaps more serious problem.

These secondary events are also of importance, but may not be the **cause** of the problem. By filtering out the less important events an Operator can concentrate on solving the real problem as quickly as possible, without the cause being hidden by a flood of events arriving at the user interface.

### **High Level Filtering** (System Administrator Task)

High level filtering takes place at the level of the **Operation Context** and Log entities. This type of event filtering also uses the discriminator construct. The discriminator construct is defined as an attribute of the Operation Context and Log entities, and contains the definition of which events are to be passed to the Alarm Handling and Event Log FMs and which events are to be blocked.

High level filtering presents the Operator with a particular view of network activity. It collects events and passes only those that match this view. Using this functionality, the user can choose to view the network on the basis of, for example, geography, equipment, severity or any other criteria defined by the user.

### **Top Level Filtering** (User task)

Top level filtering takes place at the **GUI** level. This type of filtering allows the user to quickly focus on specific information displayed. The filter criterion stays local to the current operator session.

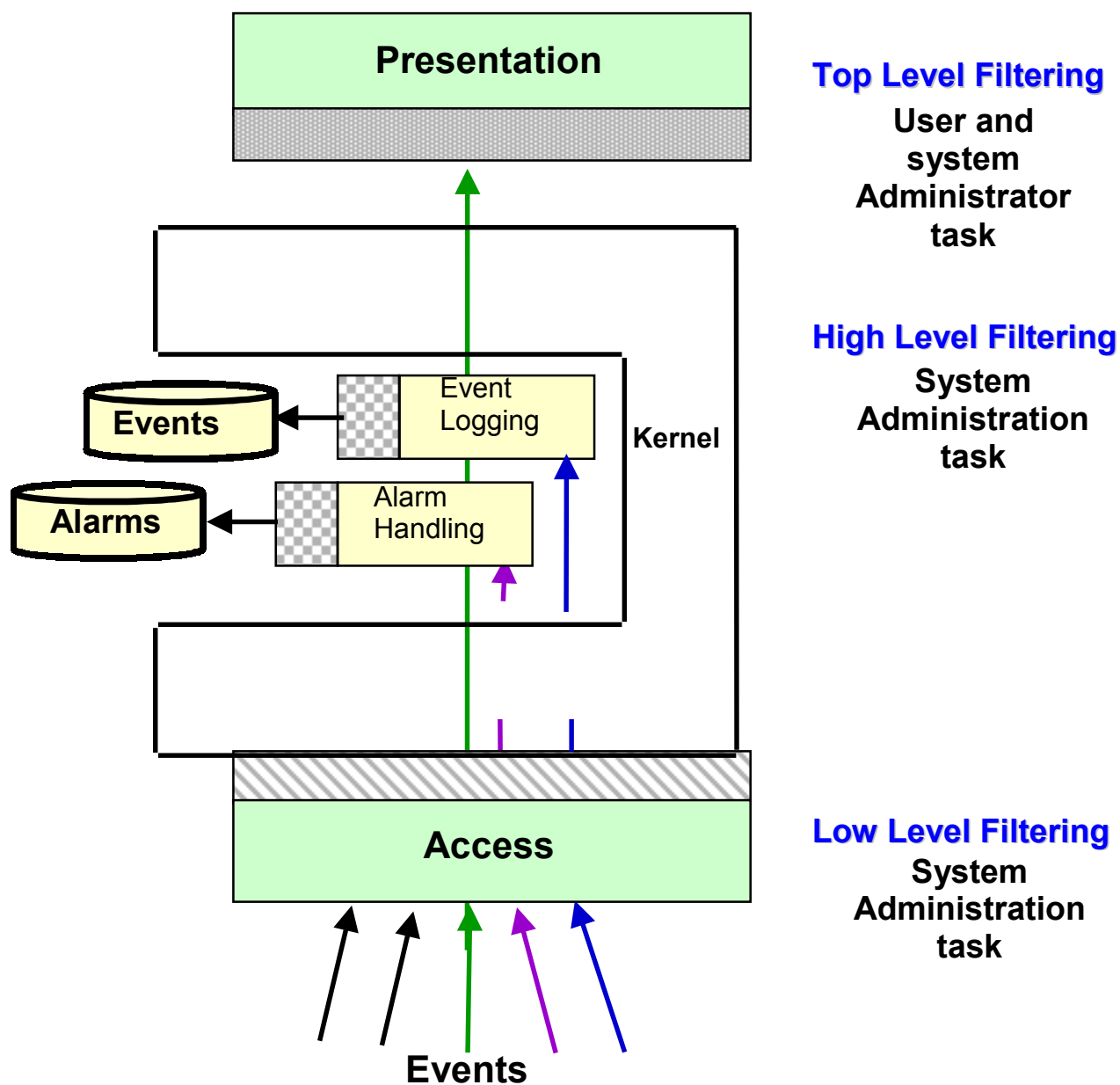


Figure 1.14 – Filtering

### Clear a Filter

If the user has applied high level filters to the alarm list and wishes now to remove those filters the user may do that by clicking the Clear button. By clicking the Clear button all alarms in the list will be viewable again.

### Top Level Filtering

By clicking the “Filter” Button on the Alarms Window screen, the User can manage what alarms are viewed. This makes the alarm list more manageable and patterns are then more visible.

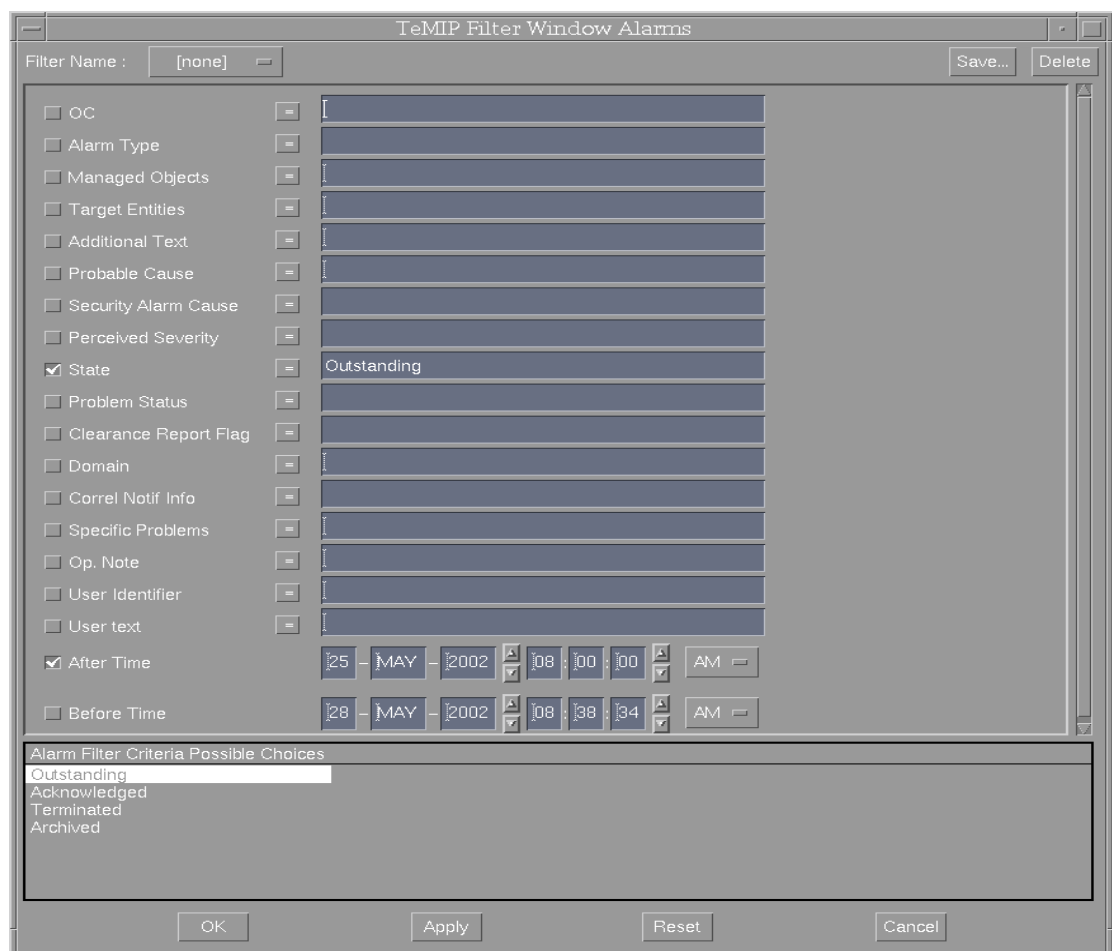


Figure 1.15 Screenshot of Top Level Filtering

## OVERVIEW OF THE ICONIC MAP WINDOW

One of the NS-T's Presentation Modules for the user interface is the Iconic Map PM. Iconic Map is also an application to NS-T. It is the direct manipulation interface that presents network entities as graphical objects on the screen, which are managed by the user by selecting and choosing operations from menus. Collections of objects are represented as maps. The user can have several maps open at once, each in its own window.

The Map window graphically displays the contents of a specific domain or the child entities of an entity. The graphical icons representing the entities can display logical and physical information. The user can select an entity in the Map window and initiate a management operation.

The Iconic Map, consisting of a geographical map (i.e. backdrop) and a dynamic foreground topology map, will show the managed objects and sub-objects that represent the managed network elements. Different level of maps can be made creating a hierarchy using the philosophy of management domain containment.

Map Items reflect changes in the network based on incoming alarm information, by changes of color, display of an information box... If, for example, the network topology is geographically based, a Map can help the operator to find the geographical location of a problem and then descend the hierarchy to find the faulty equipment itself. In this way an operator can monitor changes in the network as they happen and take action to remedy faults as soon as they occur.

The following examples of topological views could be defined:

- Overall telecommunications network view
- National telephone network interconnection view
- International telephone network view
- Individual exchange view
- Mobile network view
- Transmission network view



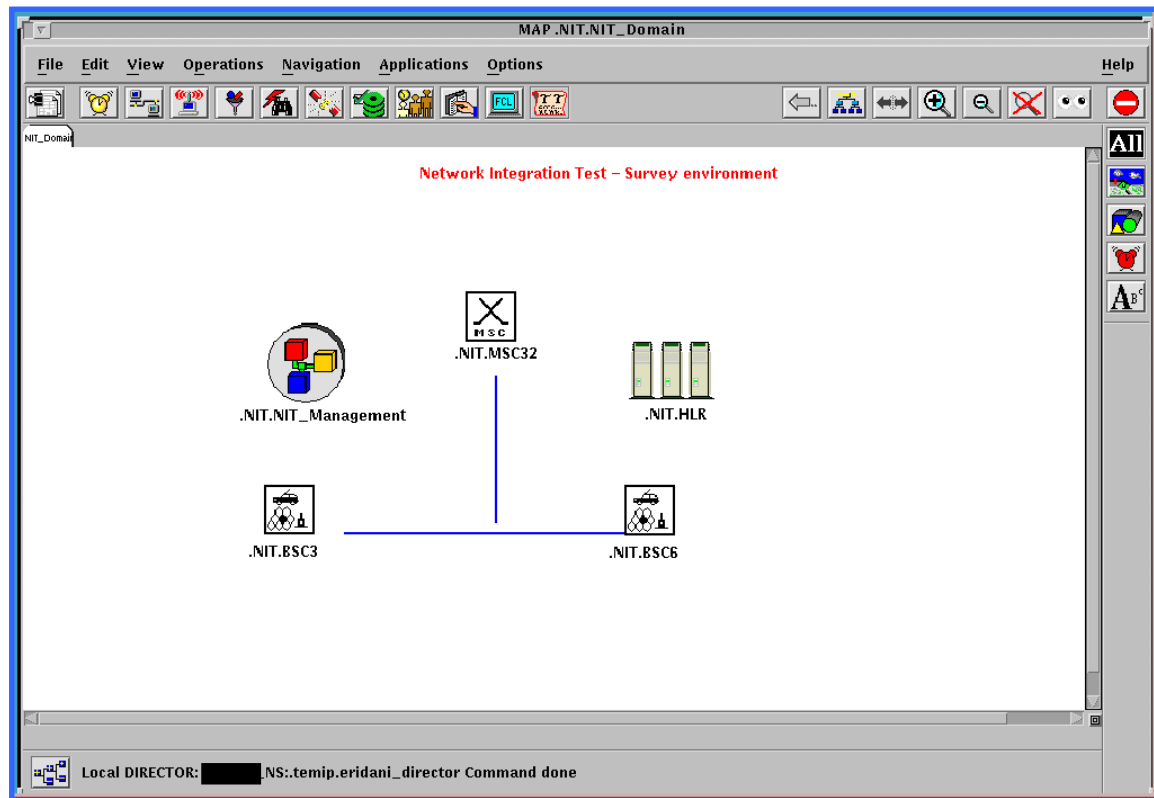
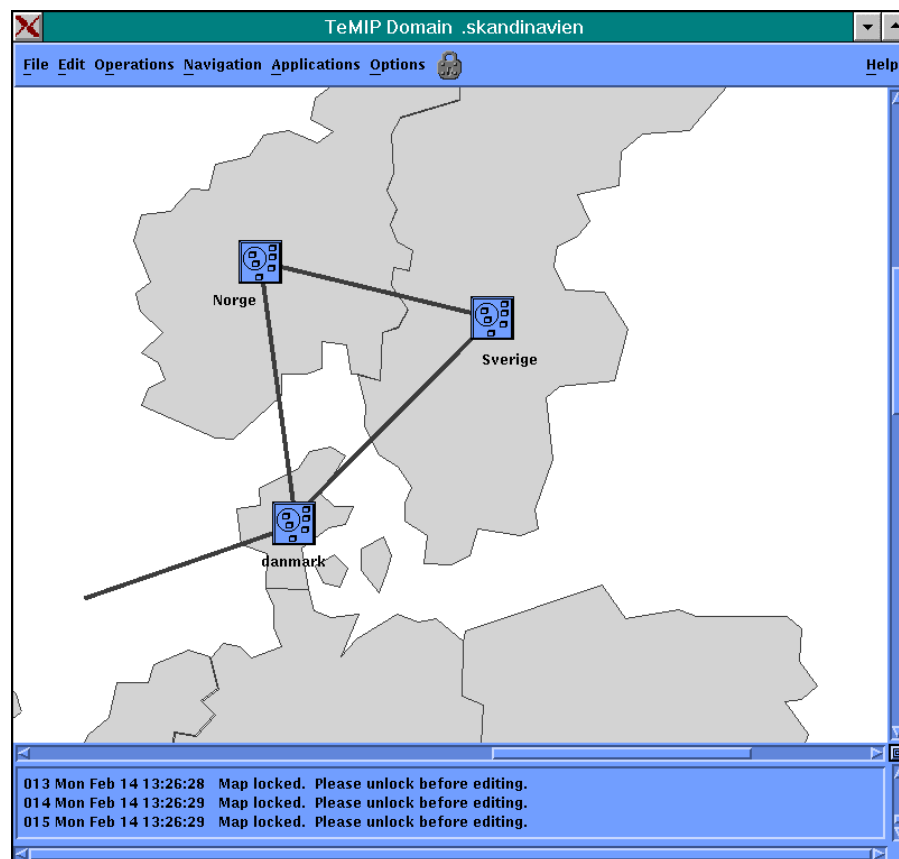


Figure 1.16 Example of an iconic map



*Figure 1.17 Example of an Iconic Map with topological background*

## HOW THE ICONIC MAP WORKS

A new alarm is reported to the Iconic Map by changing the color of the managed object icon associated with the managed network element on the topological top. A different color is used for each level of severity assigned to the alarm.

The Iconic Map PM can be used from any terminal or workstation using DECwindows/MOTIF. The Iconic Map PM uses most of the common features of any DECwindows/MOTIF application, including a title bar, a work area, a shrink-to-icon button, a maximize button, and scroll bars.

## MANAGING THE GRAPHICAL USER INTERFACE (GUI)

The GUI on the Map window allows the user to modify the contents of a Map window. The user can perform some functions on a Map window (the list presents just some of them):

- **Open an existing domain** (*relates to Alarm Surveillance*)
- Create a new domain
- **View status messages** (*relates to Alarm Surveillance*)
- Select an entity and initiate a management operation
- Configure selected operations
- Create a new Map window
- Add, modify, delete or backdrop in a Map window.

The Iconic Map allows the user to navigate up or down through Map window levels. The Iconic Map provides as well the ability to enlarge a selected area of the screen to permit increased visibility of details.

The user can work randomly on several Map windows, each window displaying different domains or entities. The navigation from one view to another is under the user choice.

### Display Navigation Aid

A Navigation Aid is available. The user can place it anywhere on the display. The portion of the display currently on the screen is indicated by a rectangle

## Iconic Map Tools

### *Title Bar*

The *Map* window title bar includes the name of the map being displayed.

### *Menu bar*

The pull-down menus give the user access to NS-T functions. An ellipsis (...) next to a menu item indicates that a dialog box will be displayed. An arrow symbol next to a menu item indicates that a submenu exists. A key sequence next to a menu item indicates an equivalent icon key combination. If nothing

appears next to a menu item, choosing the menu item initiates the function.

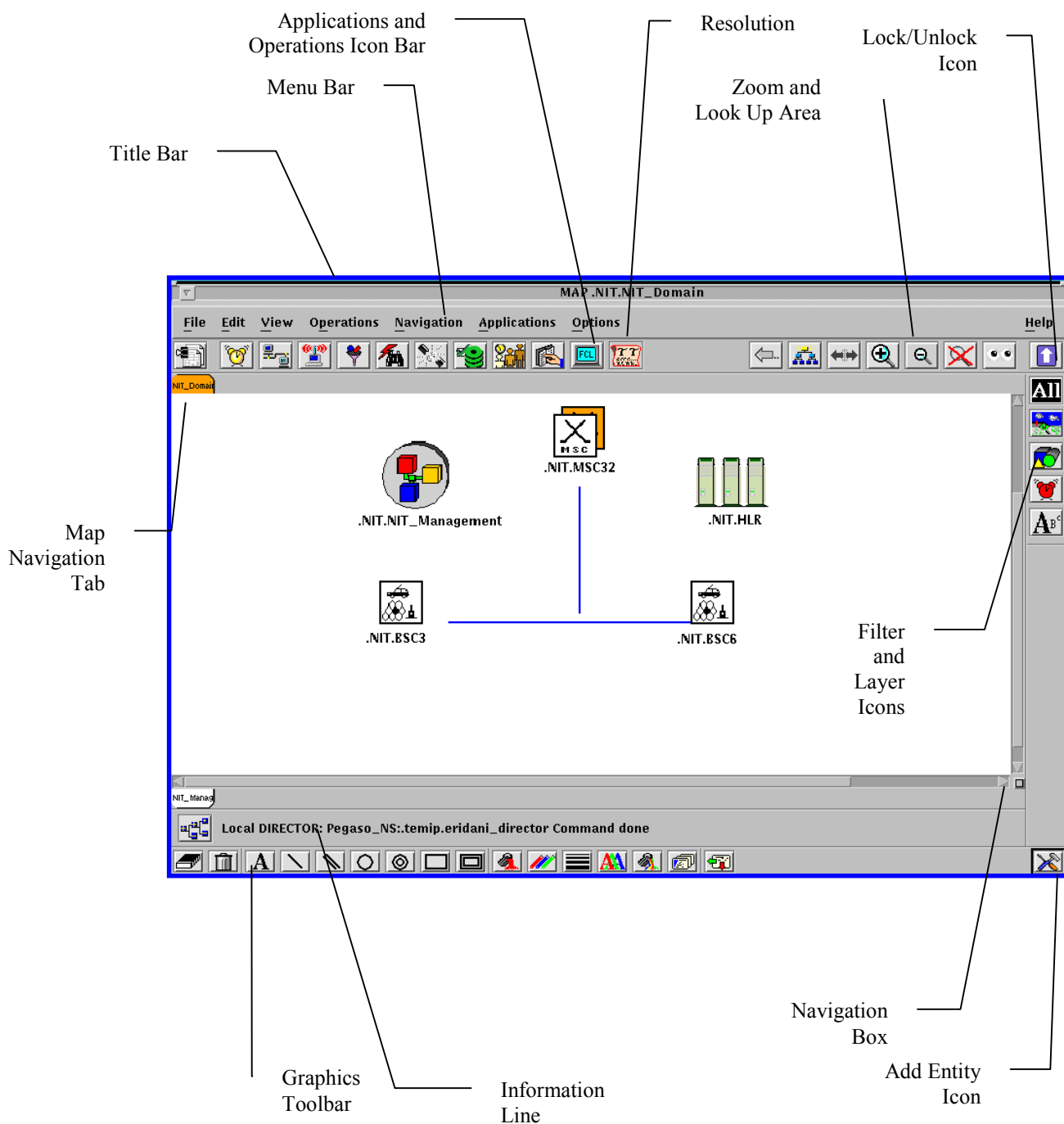


Figure 1.18 Iconic Map Tools

### *Applications and Operations Icon Bar*

This icon bar starts applications also listed in the Applications menu, and can be customized to start NS-T operations and suitably configured launched applications.

### *Map Lock Icon*

Use this icon to restrict access to the user's map. When the map is locked, the user cannot modify the current map contents, and the graphics toolbar at the bottom of the window is hidden. Users have permission to change only colors and fonts in the presentation window.

### *Top Navigation Area*

This area contains navigation tabs for the current domain and domains that are *above* the current one in the hierarchy. Use these tabs to move to a higher level and to see which domains have members with alarm notifications.

### *Zoom and Look-up Area*

This area contains zoom icons and look-up icons. The zoom icons allow the user to make all of the map fit into the *Map* window's width (Best Fit), to Zoom In and Zoom Out, to remove any zoom factor (Un-zoom), or to restore the map to the original magnification, which may have had a zoom factor applied (Restore). The navigation icons allow the user to move up one level in the tree or to the top level accessible in the tree.

### *Filter and Layer Icons*

The filter icons allow the user to filter out aspects of the *Map* window, namely, the backdrop, graphical objects, entities without alarms, and graphical texts. The user can apply several filters at once. The layer icons allow the user to look at different map layers if they have defined them. The default view is All layers. The backdrop is displayed and is the same for all layers. Entities can belong to several layers, and the user can display several layers at once.

### *Navigation Box*

Click on this box to display the *Navigation* window, which allows the user to move to parts of the map that are not displayed in the *Map* window.

### *Bottom Navigation Area*

This area shows a previously displayed child domain. A navigation tab that shows its name represents the lower-level domain or shelf view. Use this tab to return to the lower level and to see which domains have members with alarm notifications.

### *Information Line, Tree View Icon and Cancel Button*

This status line provides context-sensitive help and messages related to the NS-T operations effected. The Cancel button is displayed while an operation is in progress, and allows the user to stop the operation. The Display Tree View icon, also contained in the Information Line, allows the user to display the hierarchical structure of the domains in the *Map* window currently displayed.

### *Graphics Toolbar*

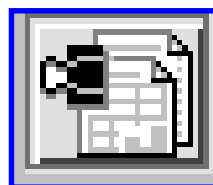
The graphics toolbar contains the graphics tools that the user can use to modify the map displayed in the *Map* window. This toolbar is only viewable when the map is unlocked and only the System Administrator should have privileges to this.

### *Add Entity Icon*

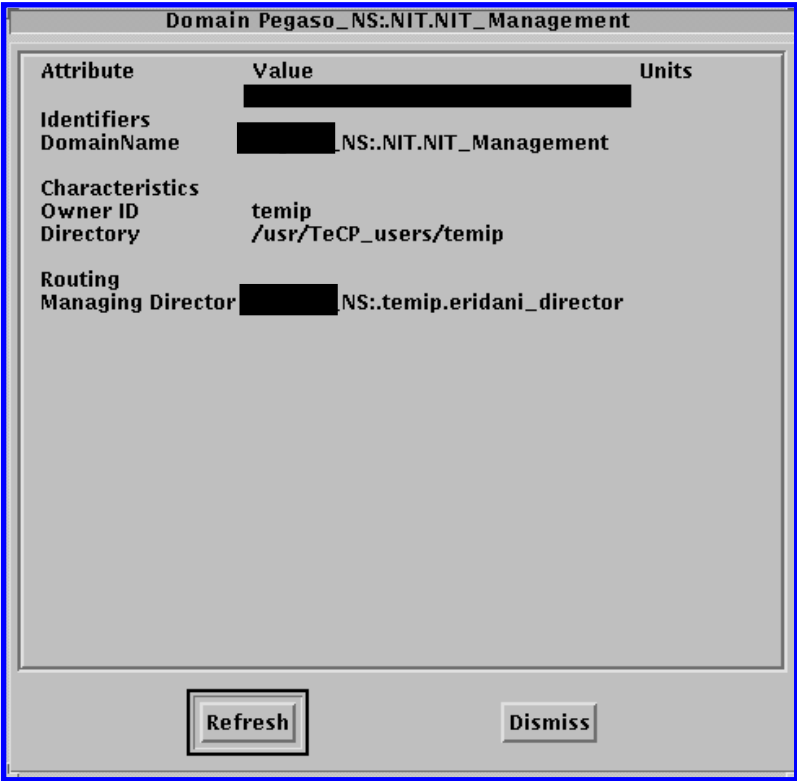
This icon gives the user quick access to the *Add Entity* window.

### *Show All Attributes*

The Show All Attributes icon is used to show all the information about an entity or icon that is selected. It is located to the left of the tool field of the Iconic Map.



*Figure 1.19a - Show all attributes icon*



The screenshot shows a window titled "Domain Pegaso\_NS::NIT.NIT\_Management". Inside, there is a table with three columns: "Attribute", "Value", and "Units". The table contains the following data:

| Attribute              | Value                                 | Units |
|------------------------|---------------------------------------|-------|
| <b>Identifiers</b>     |                                       |       |
| DomainName             | [REDACTED]_NS::NIT.NIT_Management     |       |
| <b>Characteristics</b> |                                       |       |
| Owner ID               | temip                                 |       |
| Directory              | /usr/TeCP_users/temip                 |       |
| <b>Routing</b>         |                                       |       |
| Managing Director      | [REDACTED]_NS::temip.eridani_director |       |

At the bottom of the window, there are two buttons: "Refresh" and "Dismiss".

Figure 1.19b - Show all attributes

## ASSOCIATING ALARMS TO THE ICONIC MAP

1. Select one or more Alarming Objects in the OC List view on the Iconic Map.
2. Click the Search button in the Tools toolbar

This then takes you to the Alarm Handling Window and you can see the alarms that are associated with the alarming object.

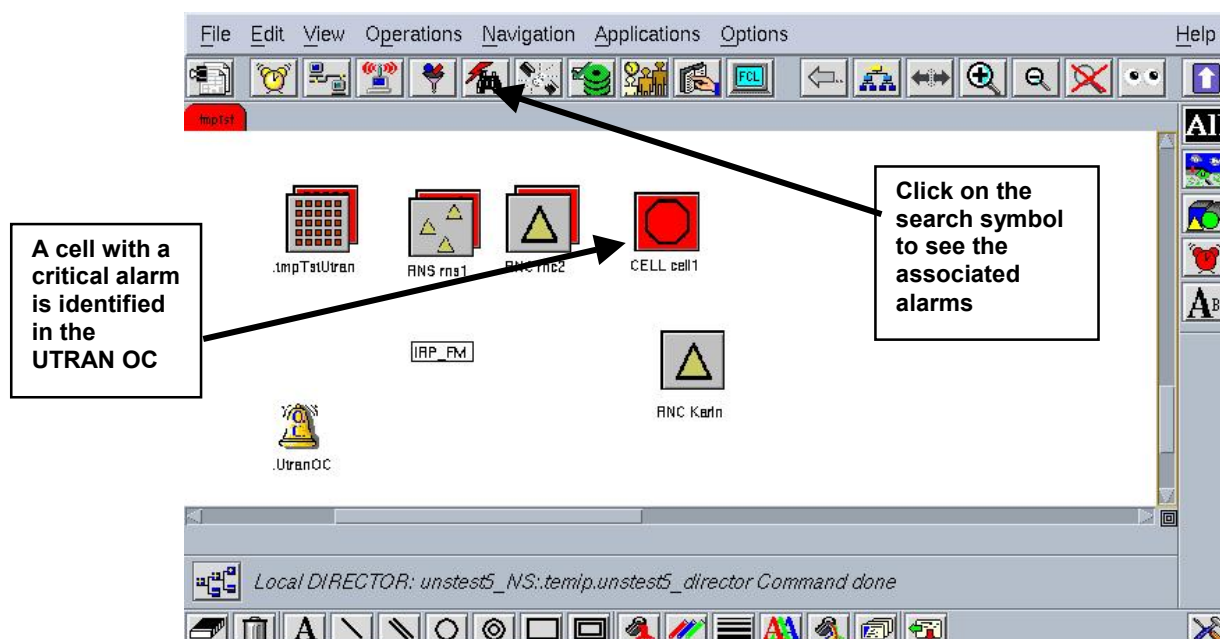


Figure 1.20 Iconic Map with a selected Alarming Object

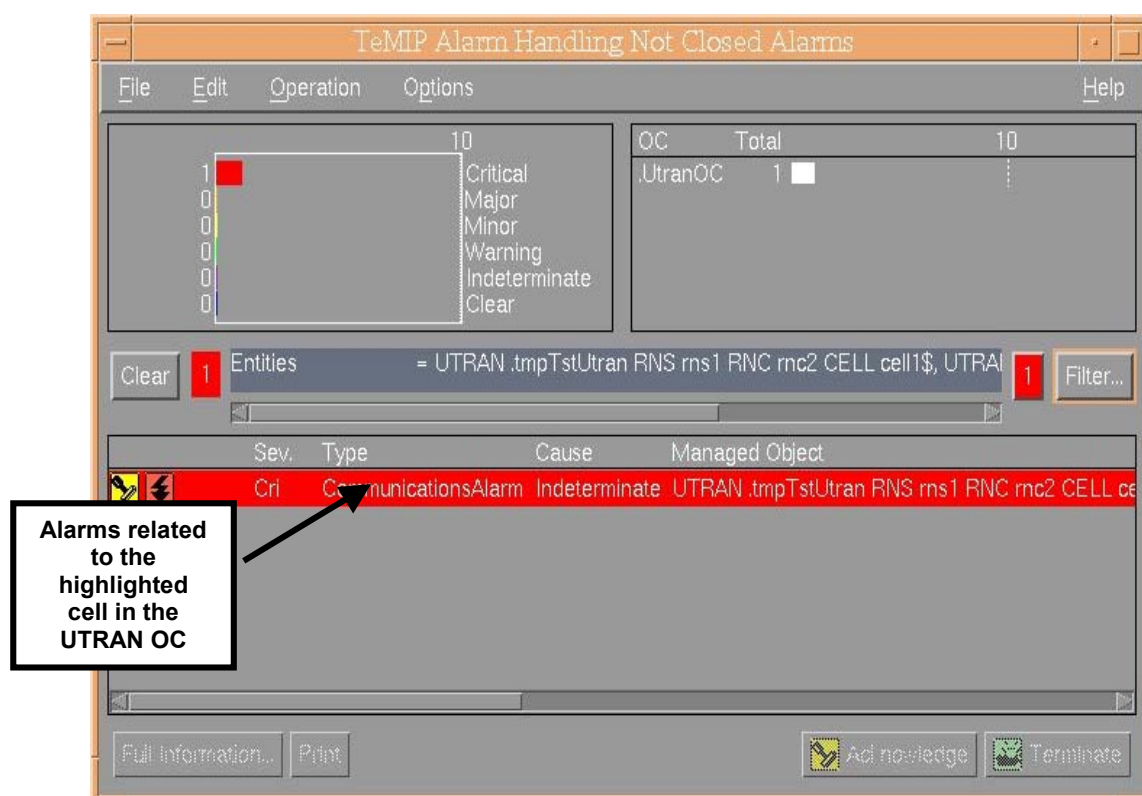


Figure 1.21 The Alarm Handling Window displaying the associated Alarms



## CLOSING ICONIC MAP WINDOWS

To close a window, choose Close Window on the File menu. This menu item is available in *Map*, *Management*, and *Add Entity* windows. The Iconic Map PM closes the window. In a *Management* window, if an operation is in progress, the Iconic Map PM first stops the operation.

In a *Map* window, if the user made changes to the map, the Iconic Map PM displays a dialog box asking if the user wants to save these changes. The user should save if it is a new item or if they have made changes since it was last saved. If the user add icons for global entities and do not save, they will not be displayed the next time the user opens the map or display the shelf view. However, entities that became members of a domain because the user placed them on its map will remain members.

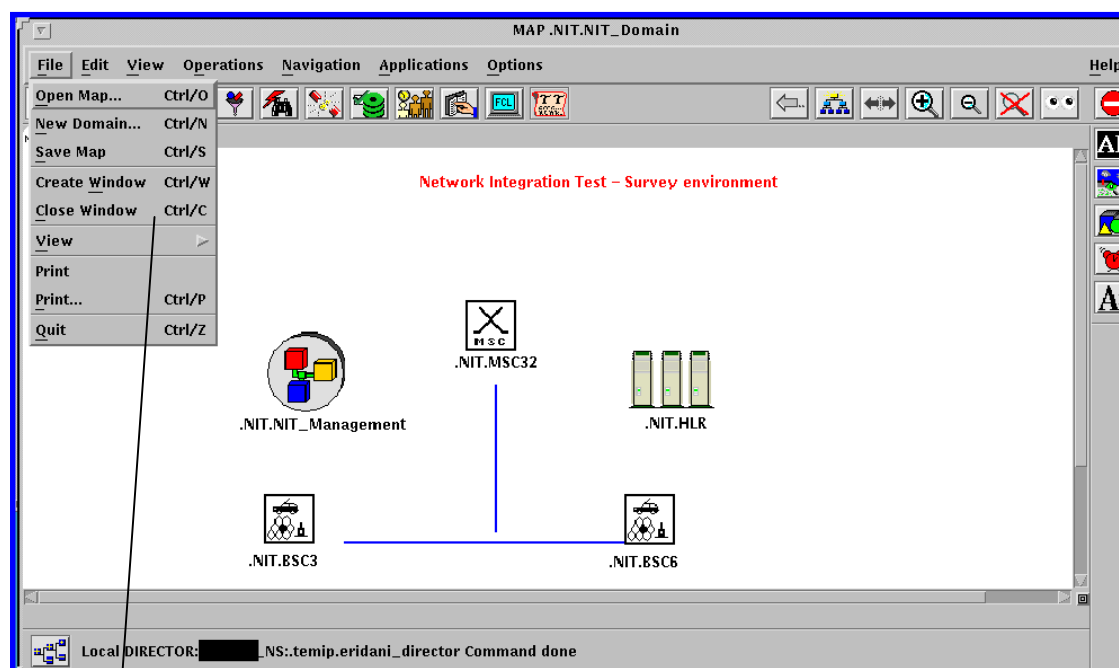


Figure 1.22 - Closing the Iconic Map Window

Click on  
Close  
Window.

## Exiting NS-T from the Iconic Map

The user can exit NS-T from any *Map* window.

When an Iconic Map session is terminated, all customization changes are lost unless the user saves the current settings before the user exit.

To exit NS-T, choose the quit menu item from the File Menu in a Map window. NS-T displays a dialog box to verify that the user wants to exit.

## ONLINE HELP

NS-T provides extensive on-line information which complement the printed documentation set. The access to help information is available at any time in any window with a help menu, without having to terminate the current session. The user can use help information, ie, concepts, rules, and instructions, at any time. When a user accesses certain types of help an internet browser will be opened and the whole documentation set will be available online. The on-line help is available from the following:

- **Iconic Map PM** - to access information the user must pull down the Help menu
- **Command Line interface** - to access information the user must enter HELP at the NS-T prompt.
- Additionally Help is available from other NS-T components with a GUI such as the Alarm Handling PM, Event Logging FM, Access Control, etc.

The context sensitive help is provided with NS-T Framework for users. This assistance could take different forms such as definitions, descriptive information, concepts, rules, instructions, examples etc. The user can add or develop their own sensitive help using Help Builder File and NS-T Framework Development Toolkit.

On the Iconic Map window the Help menu gives the access to the following categories:

- **Context Sensitive Help** - a question mark appears on the user screen. Dragging the question mark with the mouse over the item with a click on a mouse allows users to access appropriate topic in a Help window.
- **Overview** - the first screen lists the sections of the chapter, which describe the use of NS-T Framework windows and menus in the NS-T Framework Use Manual. Double click on a section reference in order to open up the topic of interest.
- **Index** - the index of the NS-T Framework Use Manual is presented in Help window. Double click on a section reference in order to open up the topic of interest.
- **Contents** - the contents page of the NS-T Framework Use manual is presented in a Help window. Double click on any item in this list opens the corresponding topic.

**Product Information** – The software product description is presented in a help window.